

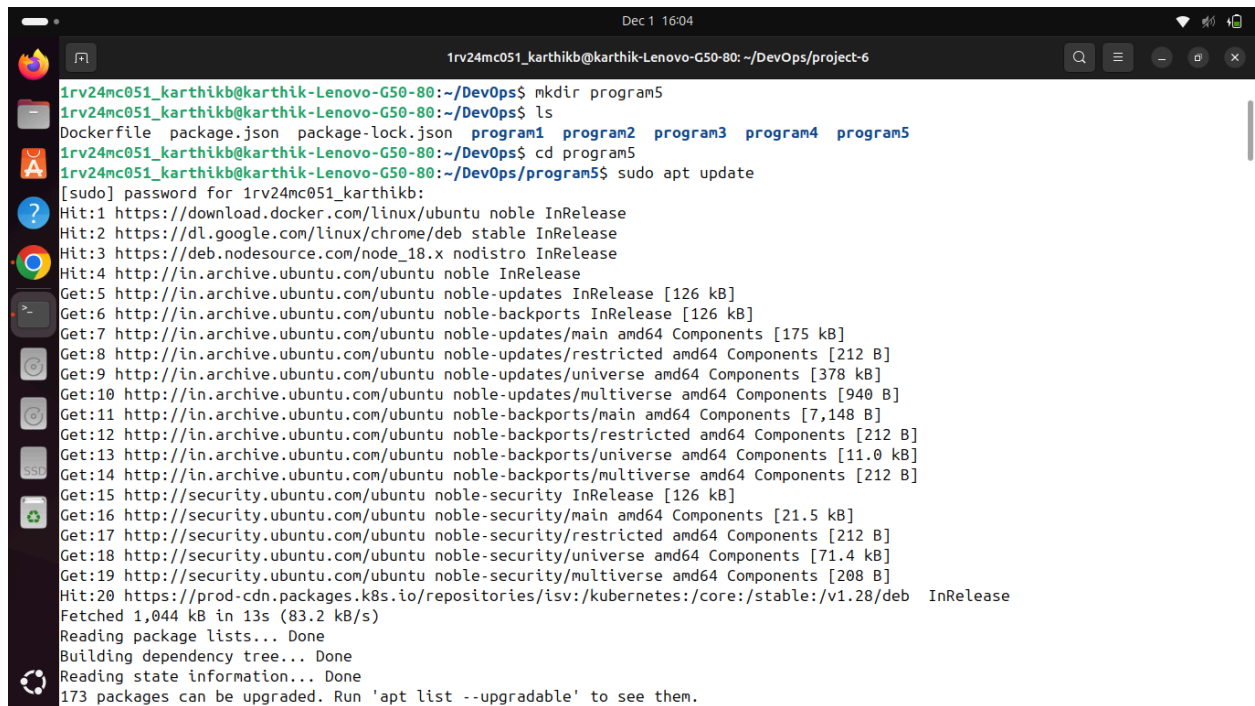
Program5

Demonstrate CI integration by running jenkins in a docker container

Project Structure

Pr5

- Dockerfile
- docker-compose.yml



```
1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/project-6
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ mkdir program5
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ ls
Dockerfile package.json package-lock.json program1 program2 program3 program4 program5
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ cd program5
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ sudo apt update
[sudo] password for 1rv24mc051_karthikb:
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,148 B]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:13 http://in.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Hit:20 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.28/deb InRelease
Fetched 1,044 kB in 13s (83.2 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 1: Create docker-compose.yml

version: '3.8'

services:

jenkins:

build: .

container_name: jenkins-ci

ports:

- "8080:8080"

- "50000:50000"

volumes:

- jenkins_home:/var/jenkins_home
- /var/run/docker.sock:/var/run/docker.sock

volumes:

Jenkins_home:

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/project-6
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ sudo curl -L "https://github.com/docker/compose/releases/download/1.29.2/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0     0      0     0      0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 12.1M 100 12.1M    0     0  267k      0  0:00:46 0:00:46 --:--:-- 549k
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ sudo chmod +x /usr/local/bin/docker-compose
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose --version
docker-compose version 1.29.2, build 5becea4c
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker --version
docker version 28.5.1, build e180ab8
docker-compose version 1.29.2, build 5becea4c
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ nano Dockerfile
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ nano docker-compose.yml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose up --build
Creating network "program5_default" with the default driver
Creating volume "program5_jenkins_home" with default driver
Building jenkins
[+] Building 2.9s (3/3) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 171B
=> ERROR [internal] load metadata for docker.io/jenkins/jenkins:lt
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
-----
> [internal] load metadata for docker.io/jenkins/jenkins:lt:
-----
Dockerfile:1
-----
1 | >>> FROM jenkins/jenkins:lt
2 |
```

Step 2: Create Dockerfile

Step 3: Build and Run Jenkins

```
Dec 1 16:05
1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/project-6
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose up --build
Building jenkins
[+] Building 438.4s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 171B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts
=> [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting image
=> => exporting layers
=> => writing image sha256:0badf7b69b672595ebdef7716be8121514dd2161ee1003d5bde3b153ce82d866
=> => naming to docker.io/library/program5_jenkins
Creating jenkins-ci ... done
Attaching to jenkins-ci
jenkins-ci | Running from: /usr/share/jenkins/jenkins.war
jenkins-ci | webroot: /var/jenkins_home/war
jenkins-ci | 2025-12-01 09:57:24.169+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
jenkins-ci | 2025-12-01 09:57:25.791+0000 [id=1] WARNING o.e.jee9.nested.ContextHandler#setContextPath: Empty contextPath
jenkins-ci | 2025-12-01 09:57:25.903+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 20
25-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm 21.0.9+10-LTS
jenkins-ci | 2025-12-01 09:57:26.848+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support
t for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
jenkins-ci | 2025-12-01 09:57:26.973+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
jenkins-ci | 2025-12-01 09:57:27.829+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /va
r/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
jenkins-ci | 2025-12-01 09:57:28.038+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHan
dler$CoreContextHandler@3c782d8e{Jenkins v2.528.2,,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreConte
xtHandler$CoreToNestedHandler@572e6fd9{STARTED}}
jenkins-ci | 2025-12-01 09:57:28.068+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7
```

Step 4: List Docker Images

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose down
Removing jenkins-ci ... done
Removing network program5_default
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose down -v
Removing network program5_default
WARNING: Network program5_default not found.
Removing volume program5_jenkins_home
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
program5_jenkins     latest      0badf7b69b67  11 minutes ago 925MB
jenkins/jenkins      lts         aa2bbbd632f3  2 weeks ago   490MB
gcr.io/k8s-minikube/kicbase v0.0.48    c6b5532e987b  2 months ago  1.31GB
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker rmi program5_jenkins:latest
Untagged: program5_jenkins:latest
Deleted: sha256:0badf7b69b672595ebdef7716be8121514dd2161ee1003d5bde3b153ce82d866
```

Step 5: Open Jenkins in Browser

```
Dec 1 15:23
1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/program5

dc2a77f462ea: Pull complete
33300af18dd0: Pull complete
c27509c3e53b: Pull complete
e4beac64dffa: Pull complete
a37b858bb47a: Pull complete
744b4792e083: Pull complete
05a7d9a8b608: Pull complete
8d2a75b252b2: Pull complete
65e4ba8066bc: Pull complete
5dc07232677a: Pull complete
7718ff514022: Pull complete
Digest: sha256:7b1c378278279c8688efd6168c25a1c2723a6bd6f0420beb5ccfabee3cc3bb1
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program5$ docker-compose up --build
Building jenkins
[+] Building 211.5s (4/5)
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 171B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts
=> [2/2] RUN apt-get update && apt-get install -y docker.io
=> => Ign:8 http://deb.debian.org/debian trixie/main amd64 dbus-session-bus-common all 1.16.2-2
=> => # Get:9 http://deb.debian.org/debian trixie/main amd64 dbus-daemon amd64 1.16.2-2 [159 kB]
=> => # Get:10 http://deb.debian.org/debian trixie/main amd64 dbus-system-bus-common all 1.16.2-2 [53.5 kB]
=> => # Get:11 http://deb.debian.org/debian trixie/main amd64 dbus amd64 1.16.2-2 [71.6 kB]
=> => # Get:12 http://deb.debian.org/debian trixie/main amd64 runc amd64 1.1.15+ds1-2+b4 [3230 kB]
=> => # Get:13 http://deb.debian.org/debian trixie/main amd64 containerd amd64 1.7.24~ds1-6+b4 [32.8 MB]

docker:default
0.0s
0.0s
0.0s
0.1s
0.0s
0.5s
210.8s
```